

SMITH ISLAND, WA, USA

WMO: 994180

Lat: 48.3206N

Lon: 122.8310W

Elev: 50

StdP: 14.67

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	30.1	33.3	N/A	N/A	N/A	N/A	N/A	N/A	48.6	45.8	44.6	45.7	14.1	20	0.722

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.3	65.9	N/A	63.3	N/A	61.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6.7	270

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Extreme Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		DB	WB
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
43.0	36.7	31.9	27.4	73.5	4.9	3.0	23.9	75.6	21.0	77.4	18.3	79.0	14.8	81.2		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	43.3	43.3	45.6	48.3	51.4	54.4	56.6	57.0	55.6	51.1	46.2	43.1
	DBStd	6.10	4.42	4.01	3.46	2.75	2.51	2.54	2.72	2.30	2.31	2.71	4.02	4.39
	HDD50	968	208	187	142	63	13	0	0	0	0	19	121	215
	HDD65	5584	672	607	603	501	420	319	262	248	281	429	563	679
	CDD50	859	1	0	4	12	58	131	203	217	170	54	8	1
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH74	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	12.7	14.8	14.2	14.0	12.9	11.9	11.9	11.6	10.4	9.1	11.1	14.9	15.7
	PrecAvg	24.6	3.0	2.0	2.3	1.9	1.8	1.4	0.8	0.9	1.1	2.4	3.8	3.1
Precipitation	PrecMax	30.6	4.5	4.6	3.2	3.2	3.5	2.4	2.1	3.1	2.6	4.5	7.6	5.4
	PrecMin	18.7	1.1	0.3	1.0	0.3	0.4	0.2	0.1	0.1	0.1	0.7	1.1	1.3
	PrecStd	3.5	0.8	0.8	0.7	0.6	0.8	0.6	0.5	0.7	0.7	1.0	1.6	1.0
	PrecStd	3.5	0.8	0.8	0.7	0.6	0.8	0.6	0.5	0.7	0.7	1.0	1.6	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	53.2	56.7	58.3	62.0	68.4	71.1	68.8	65.5	60.3	56.4	54.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	51.4	50.7	53.0	55.2	58.6	63.1	66.2	64.7	62.9	57.6	54.2	51.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	50.1	49.3	51.2	53.4	56.6	59.6	62.7	62.5	60.9	56.1	52.6	49.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	48.7	48.0	49.9	51.9	55.0	57.6	60.3	60.6	59.4	54.7	51.1	48.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.4	6.4	6.7	6.6	7.2	8.1	8.3	7.7	5.6	5.2	4.7
		MCDBR	6.1	6.2	7.8	8.7	8.8	10.3	12.1	11.2	10.1	7.1	6.1	5.7
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.318	0.316	0.328	0.350	0.374	0.355	0.346	0.351	0.339	0.329	0.323	0.314
	taud		2.531	2.554	2.536	2.467	2.415	2.488	2.544	2.530	2.561	2.575	2.526	2.497
	Ebn at Noon		242	271	284	286	282	287	289	283	277	262	237	226
	Edh at Noon		19	23	27	32	36	33	31	30	27	23	19	18
All-Sky Solar Radiation	RadAvg		298	578	897	1353	1727	1888	2068	1748	1249	689	356	240
	RadStd		40	77	105	126	136	180	146	114	104	75	53	32

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page